Evolution of Cosmic Structure

Lecture 18
Galaxy groups and clusters: Scaling relations

Suhail Dhawan

UNIVERSITY OF BIRMINGHAM

s.dhawan@bham.ac.uk

Learning Outcomes

What are the contents and key properties of the largest collapsed structures in the Universe at the present time?

How do galaxies in clusters differ from the field, and what environmental processes are at work?

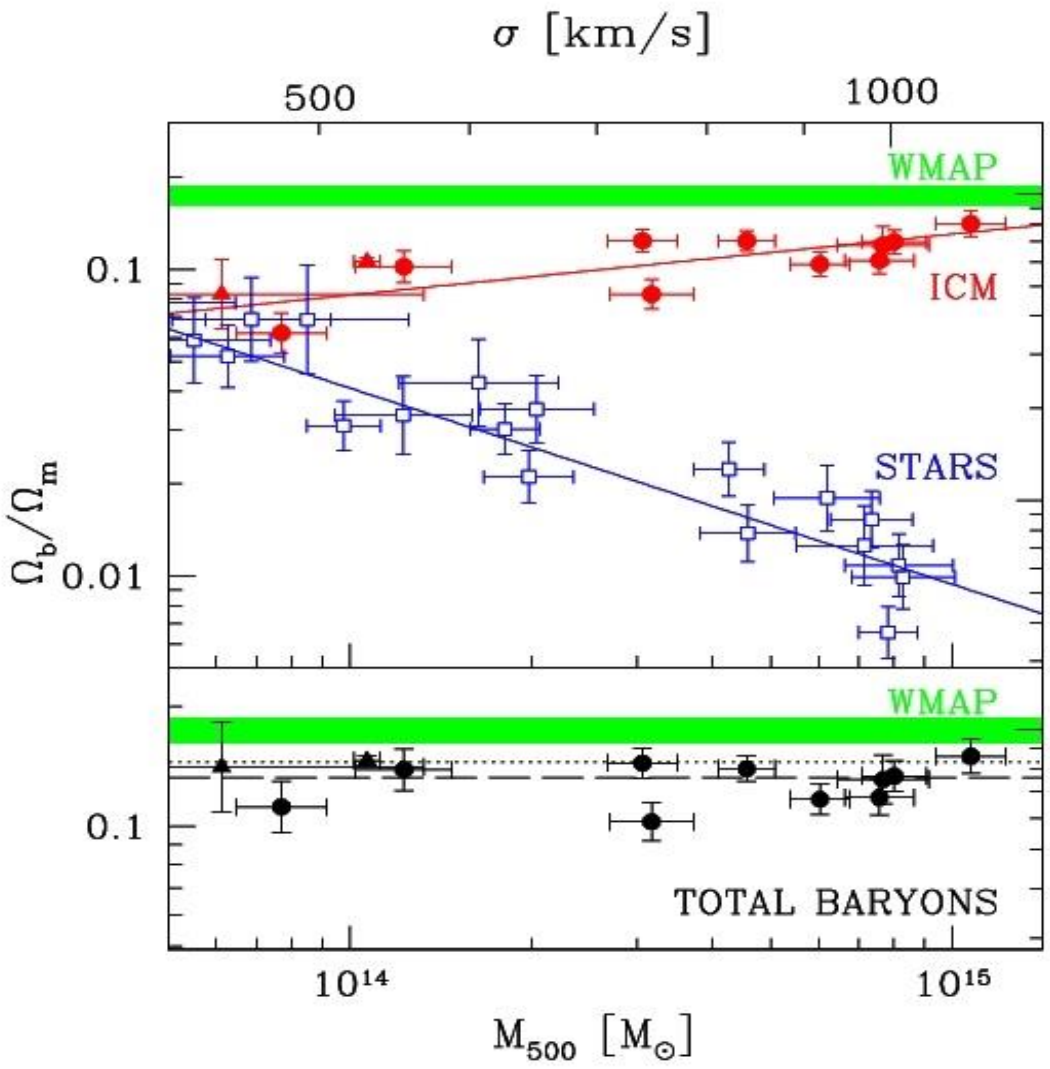
What do we learn from the properties of the hot intergalactic gas?

What scaling properties do groups and clusters follow, and how can they be understood?

How can the dominant dark matter be traced?

- Global properties and how do we find groups/clusters?
- Galaxies in clusters
- Intergalactic gas and scaling properties
- Dark matter
- Clusters and Cosmology

Global properties



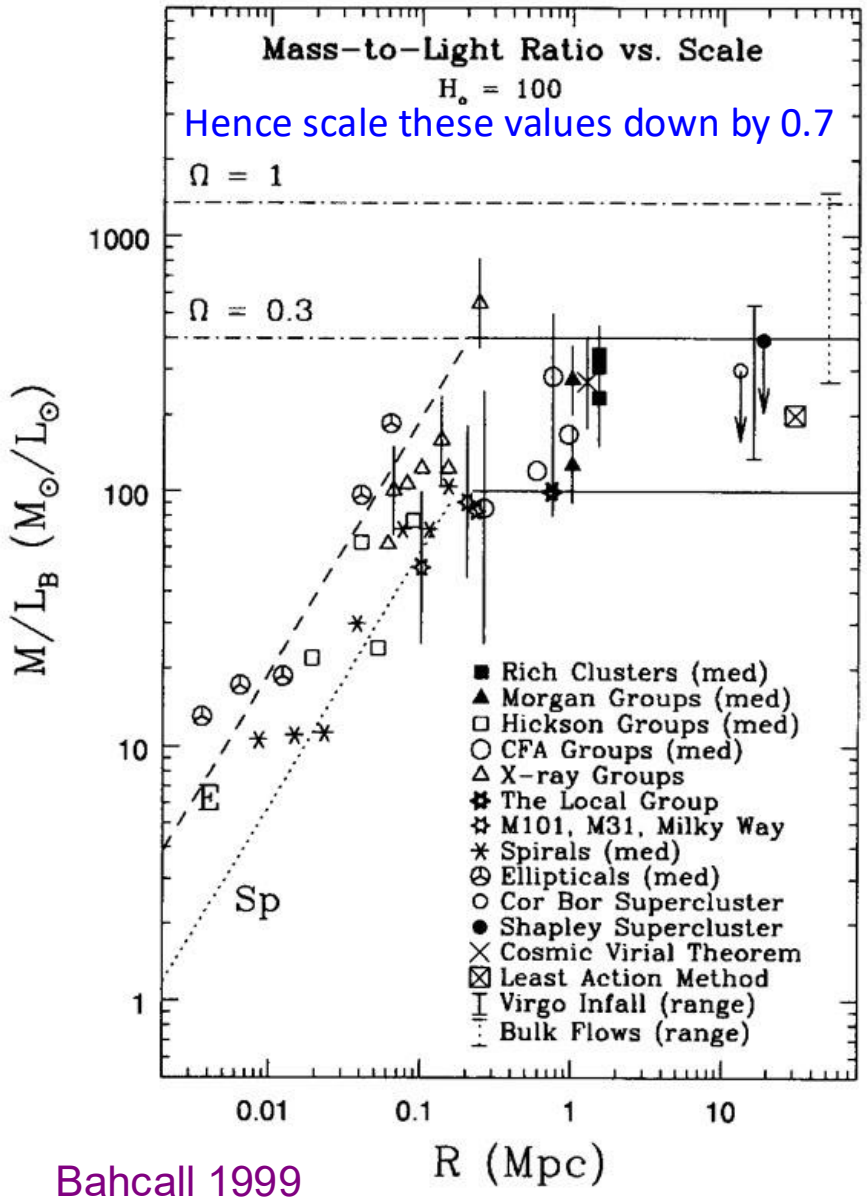
Gonzalez et al 2007

The balance between stars and gas is observed to vary substantially from groups to clusters.

The hot gas fraction rises with system mass, whilst the stellar mass fraction falls. In poor groups, the two are approximately equal.

The *total* baryon fraction seems to be rather less than the cosmic mean value of 17% (from WMAP). The issue of whether it drops in groups is still controversial.

Global properties



Bahcall 1999

The mass-to-light ratio (in the blue band) is observed to be ~200 $M_\odot/L_\odot$ in rich clusters. This can be compared with the mean value inferred for the Universe as a whole.

The mean blue band luminosity density of the Universe derived by integrating over the 2dF galaxy LF is

$$j_B = 1.7 \times 10^8\, h_{70}\, L_\odot\ \mathrm{Mpc}^{-3}$$

and the critical density of the Universe is equivalent to $\rho_c = 1.35 \times 10^{11}\, h_{70}^2\, M_\odot\ \mathrm{Mpc}^{-3}$.

So, for $\Omega_m = 0.28$, we have a cosmic M/L of $0.28 \times 1.35 \times 10^{11}\, h_{70} / 1.7 \times 10^8 = 222\, h_{70}$, in solar units.

Gas in groups and clusters

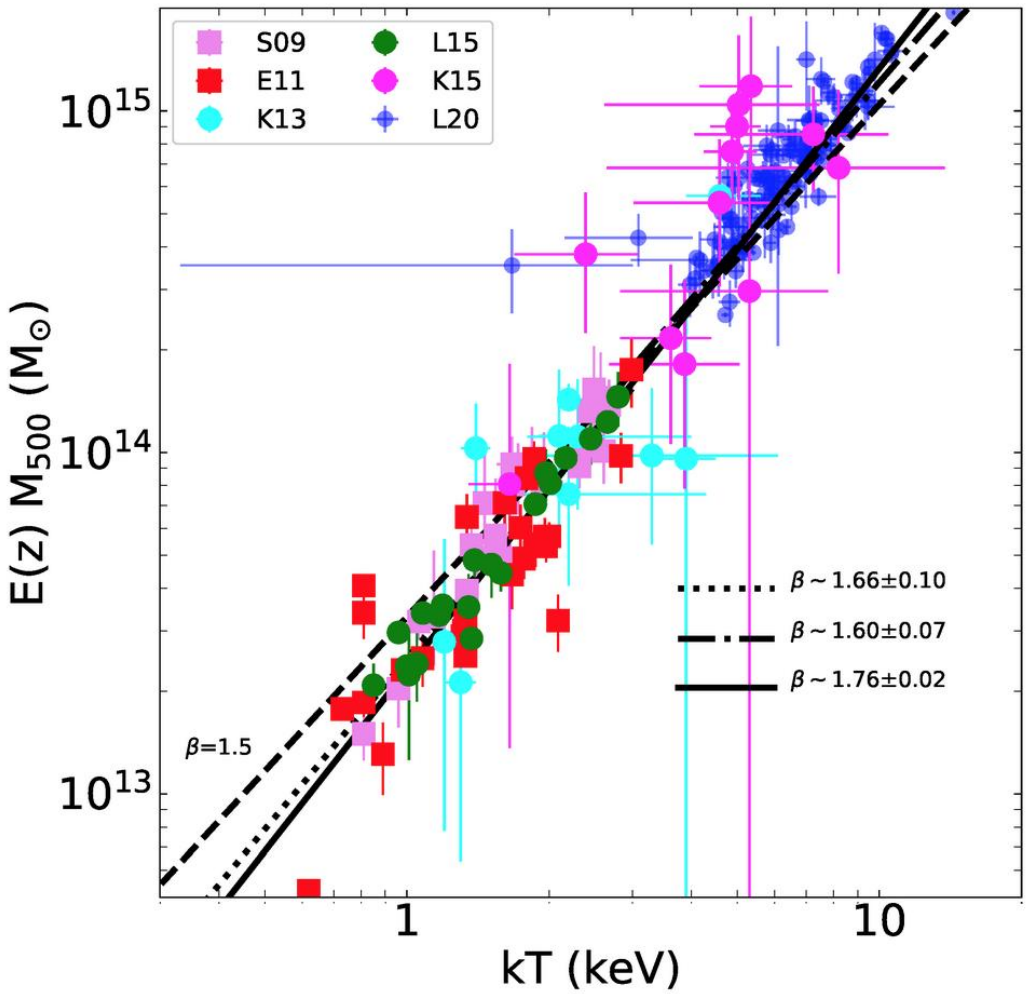
We saw earlier in this section that the VT implies a connection between the mass and radius of a cluster, and the velocity dispersion of its galaxies, of the form

$$M = \frac{3\sigma^2 R}{\eta G}.$$

From the top hat collapse model: all virialised systems are expected to have a mean density, within their virialised region of approximately $200\rho_c$ (where ρ_c is the critical density of the Universe at the time they virialise).

Hence for clusters which virialise at the same time, their mass and radius must be simply related by $M \propto R^3$.

It follows that $M^{2/3} \propto \sigma^2 \propto T$, and so we expect cluster mass to scale as $T^{3/2}$, which is close to what is observed.



Lovisari et al 2021

N.B. 1 keV $= 1.16 \times 10^7$K.

Gas in groups and clusters

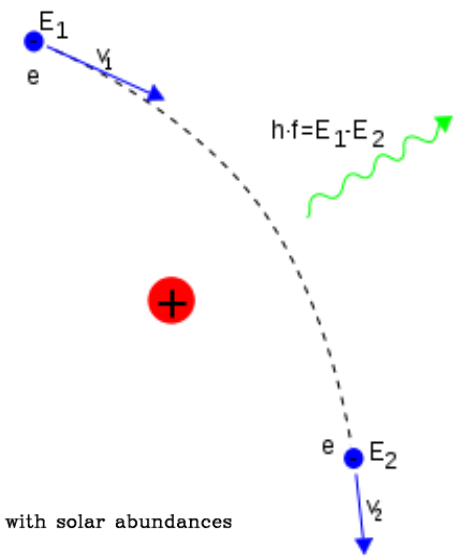
The hot gas radiates through a combination of bremsstrahlung and atomic line emission.

The bremsstrahlung emissivity has the form:

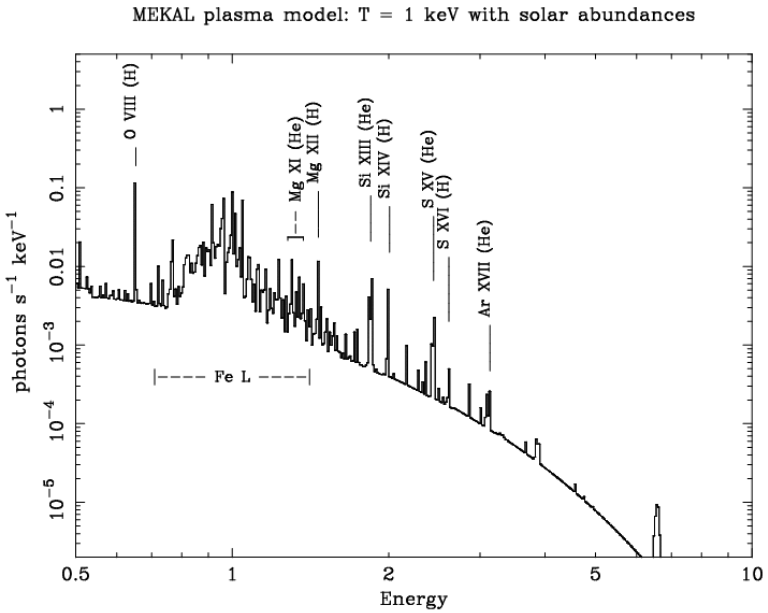
$$\epsilon(E) = A\left(\sum_i Z_i^2 n_e n_i\right) T^{1/2} g(T,E) e^{-E/kT},$$

where for gas of temperature T, the Gaunt factor is a slowly varying function of energy, which can be approximated as

$$g(T,E) \propto E^{-0.4}$$



Emission from bound-bound transitions becomes increasingly important for gas with T<3 keV, and line features within X-ray spectra are indicative of individual ionic species. Various codes (Mekal, APEC etc) are available to calculate the spectrum emerging from hot plasmas in collisional ionisation equilibrium.



Gas in groups and clusters

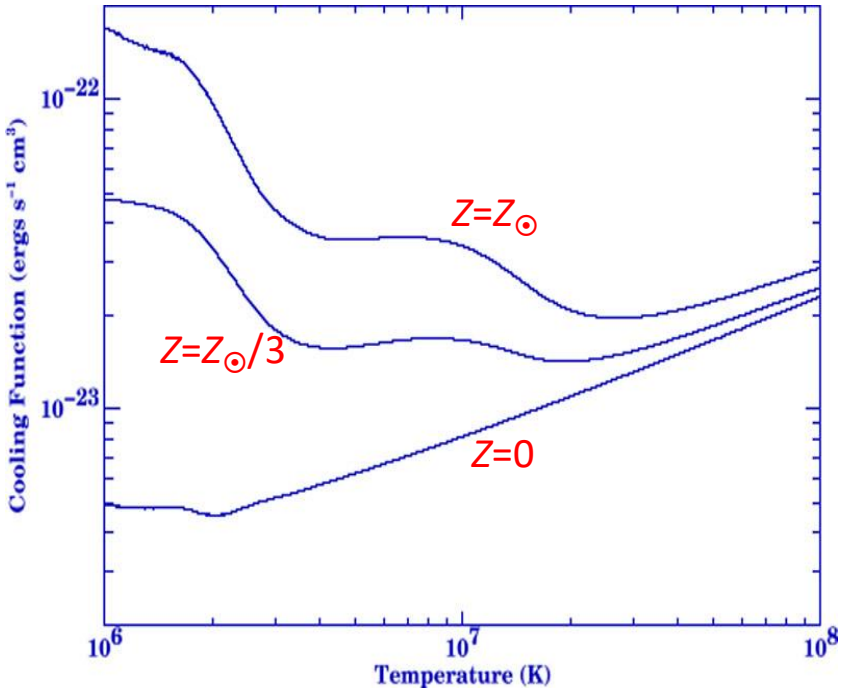
In practice, the spectral resolution of X-ray spectrometers is limited, so that lines are blended together. However, it is still possible to fit hot plasma models to constrain the temperature and metallicity of the hot gas.

The total emissivity (power radiated per unit volume) of the gas can be represented as

$$\epsilon = n_e n_H \Lambda(T, Z)$$

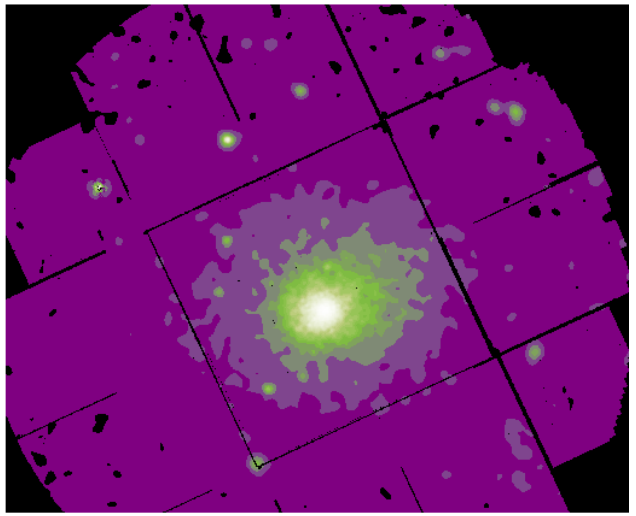
where $\Lambda(T, Z)$ is the cooling function of gas of temperature *T* and metallicity *Z*.

$\Lambda(T) \propto T^{1/2}$ for bremsstrahlung, which dominates for gas with *T*>3 keV. However, for cooler gas with significant metal content, line emission causes $\Lambda(T)$ to flatten, and even to *rise* towards lower temperatures.



X-ray mass analysis

The distribution of the X-ray emission form the hot gas in a cluster can be used to derive not only the distribution of hot gas, but also that of the total gravitating mass, and hence the dark matter. The basic approach is to use the equation of hydrostatic equilibrium $\frac{dP}{dr} = -\frac{GM(<r)\rho}{r^2}$ to infer the mass profile from the pressure gradient.



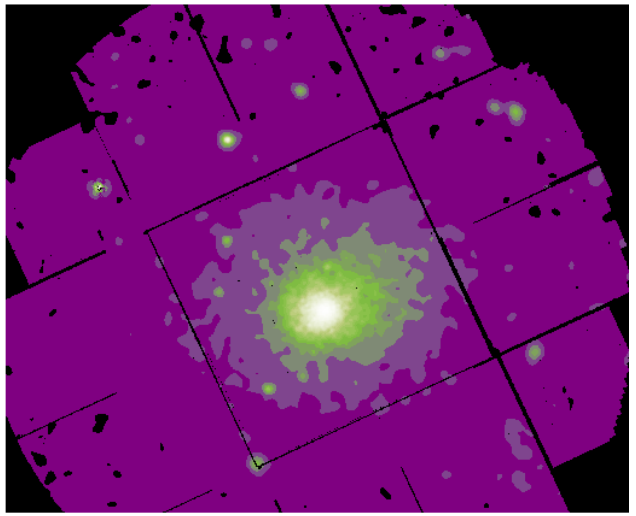
X-ray mass analysis

The distribution of the X-ray emission form the hot gas in a cluster can be used to derive not only the distribution of hot gas, but also that of the total gravitating mass, and hence the dark matter. The basic approach is to use the equation of hydrostatic equilibrium $\frac{dP}{dr} = -\frac{GM(<r)\rho}{r^2}$ to infer the mass profile from the pressure gradient.

The ideal gas law is: $$P = \frac{\rho k T}{\mu m_H}$$

Inserting the ideal gas law into the equation for hydrostatic equilibrium and differentiating gives:

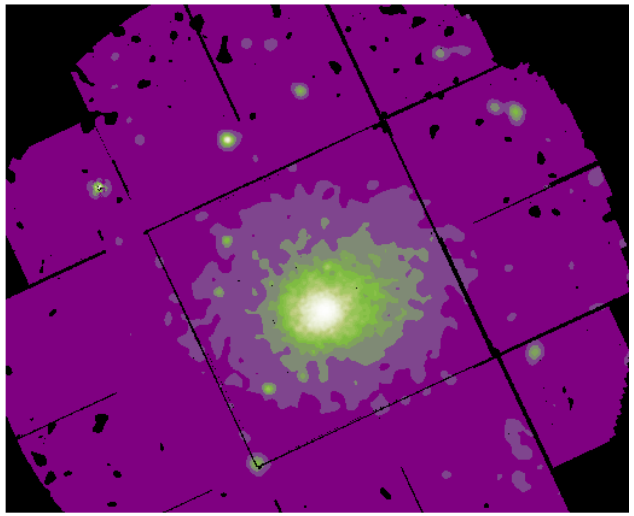
$$\frac{d}{dr}\left(\frac{\rho k T}{\mu m_H}\right) = \frac{k}{\mu m_H}\left[\rho\frac{dT}{dr} + T\frac{d\rho}{dr}\right] = -\frac{GM(r)}{r^2}\rho$$



X-ray mass analysis

The distribution of the X-ray emission form the hot gas in a cluster can be used to derive not only the distribution of hot gas, but also that of the total gravitating mass, and hence the dark matter. The basic approach is to use the equation of hydrostatic equilibrium $\frac{dP}{dr} = -\frac{GM(<r)\rho}{r^2}$ to infer the mass profile from the pressure gradient.

The ideal gas law is: $$P = \frac{\rho k T}{\mu m_H}$$



Inserting the ideal gas law into the equation for hydrostatic equilibrium and differentiating gives:

$$\frac{d}{dr}\left(\frac{\rho k T}{\mu m_H}\right) = \frac{k}{\mu m_H}\left[\rho\frac{dT}{dr} + T\frac{d\rho}{dr}\right] = -\frac{GM(r)}{r^2}\rho$$

Factoring out the T and ρ and multiplying by r gives the expression:

$$\frac{kT\rho}{\mu m_H}\left[\frac{r}{T}\frac{dT}{dr} + \frac{r}{\rho}\frac{d\rho}{dr}\right] = -\frac{GM(r)}{r}\rho$$

X-ray mass analysis

The distribution of the X-ray emission form the hot gas in a cluster can be used to derive not only the distribution of hot gas, but also that of the total gravitating mass, and hence the dark matter. The basic approach is to use the equation of hydrostatic equilibrium $\frac{dP}{dr} = -\frac{GM(<r)\rho}{r^2}$ to infer the mass profile from the pressure gradient.

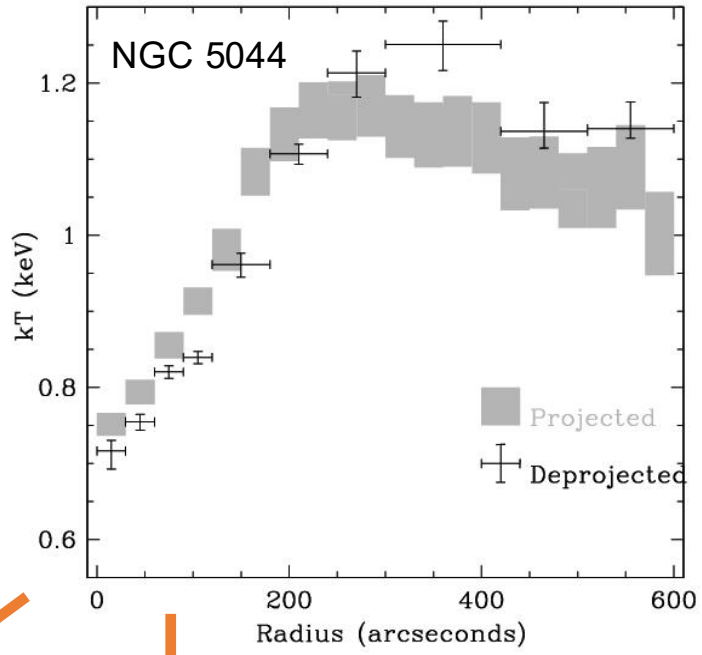
Rearranging the expression:

$$M(r) = -\frac{k_{\rm B}Tr}{\mu m_{\rm H}G}\left[\frac{{\rm dln}\rho_{\rm gas}}{{\rm dln}r} + \frac{{\rm dln}T}{{\rm dln}r}\right]$$

X-ray mass analysis

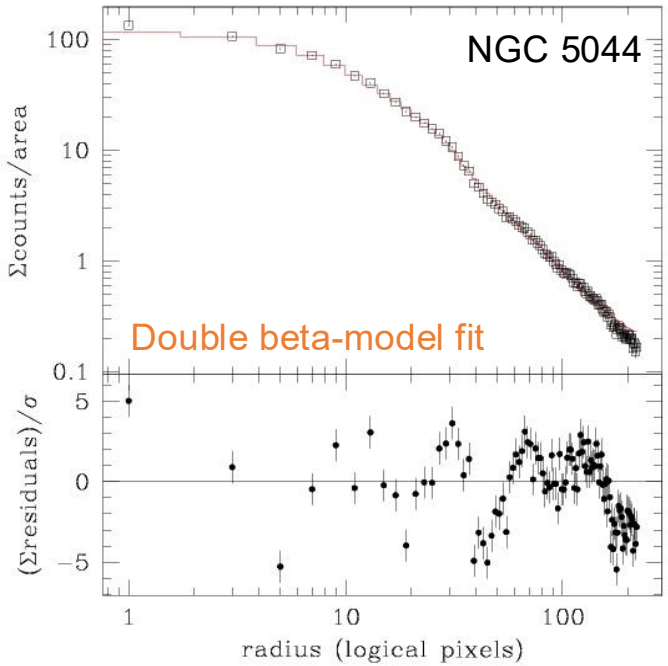
Temperature as a function of 3D radius in the system is typically derived from a deprojected fit to spectra from a set of concentric annuli.

A smooth function may then be fitted through the deprojected *T* values, to give *T(r),* and allow the log derivative to be calculated.



$$M(r) = -\frac{k_{\rm B}Tr}{\mu m_{\rm H}G}\left[\frac{{\rm dln}\rho_{\rm gas}}{{\rm dln}r} + \frac{{\rm dln}T}{{\rm dln}r}\right]$$

X-ray mass analysis



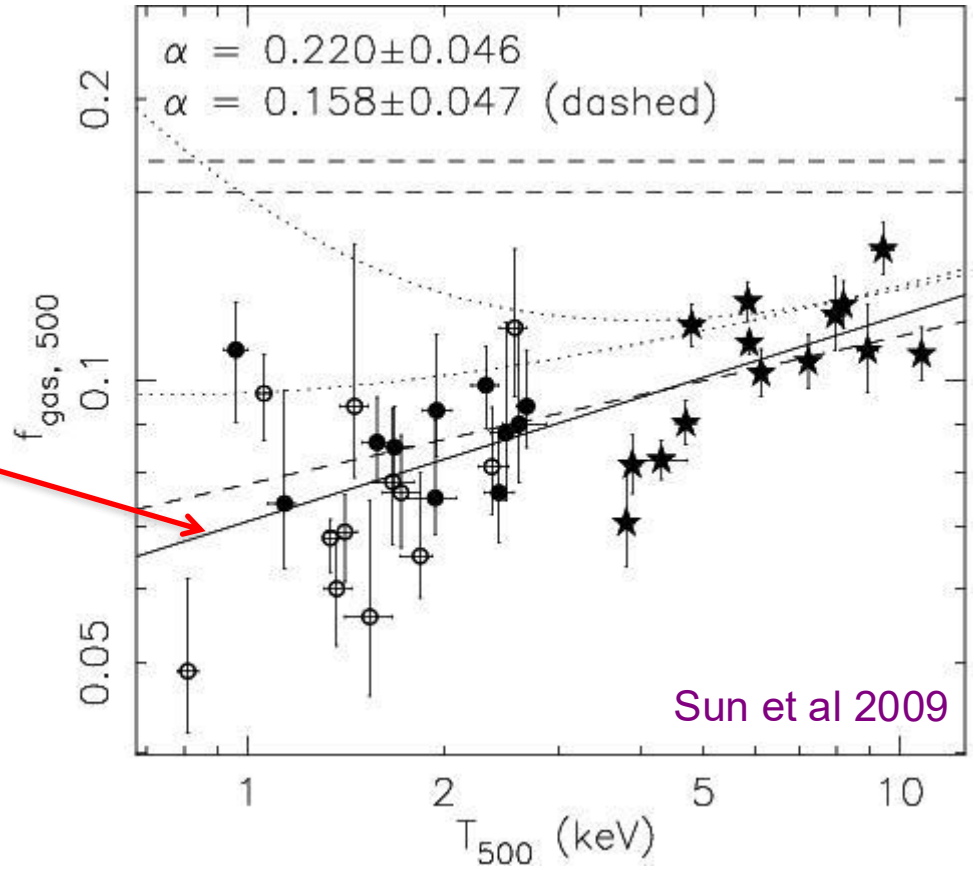
Gas density may be derived by fitting the deprojected densities inferred from the the spectral fits. For better resolution, it can be derived from a fit to the X-ray surface brightness profile, with correction for $\Lambda(T,Z)$ if necessary.

$$M(r) = -\frac{k_{\rm B}Tr}{\mu m_{\rm H}G}\left[\frac{{\rm dln}\rho_{\rm gas}}{{\rm dln}r} + \frac{{\rm dln}T}{{\rm dln}r}\right]$$

X-ray mass analysis

Combining the results for total and gas masses enables the gas fractions for groups and clusters to be derived. This rises with cluster mass. The best analysis (Sun et al. 2009) derived a trend

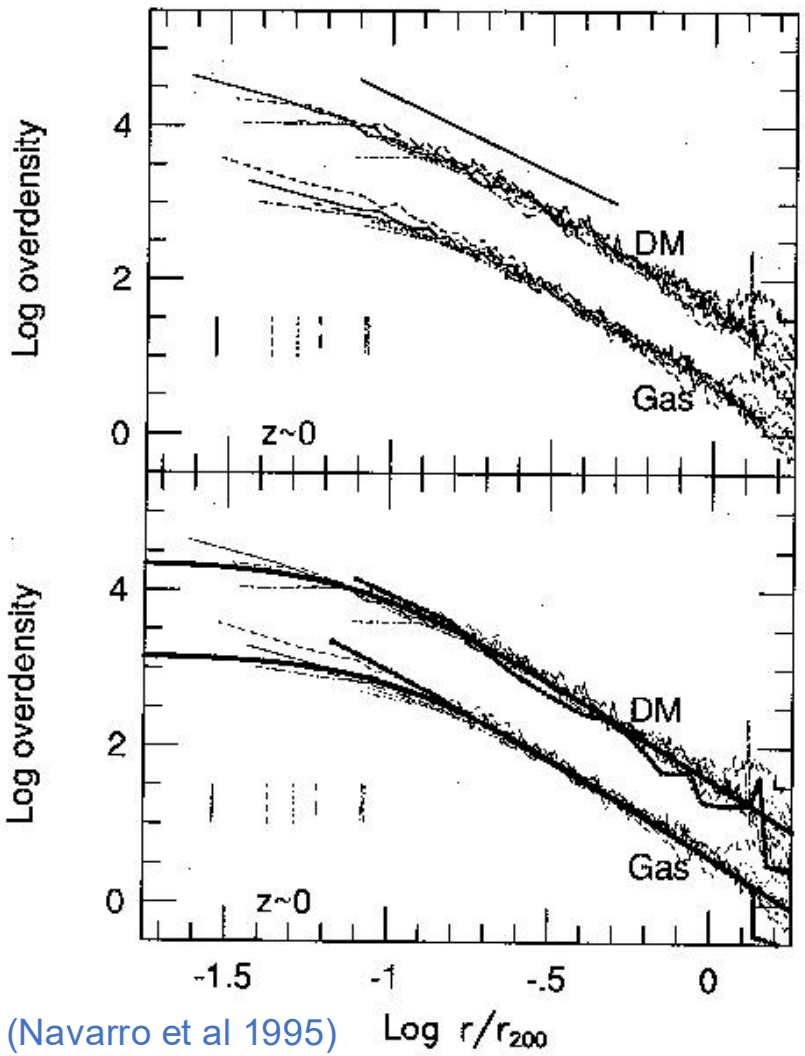
$$f_{gas} = 0.0708\ T^{0.22}$$



X-ray scaling properties

Cosmological simulations including gravity and simple gas physics produce dark halos which are almost self-similar, when scaled to a radius enclosing fixed overdensity (e.g. r_{200}).

Also, gas tracks dark matter within these halos. This behaviour would generate clusters with well-defined X-ray scaling relations.



(Navarro et al 1995)

X-ray scaling properties

Cosmological simulations including gravity and simple gas physics produce dark halos which are almost self-similar, when scaled to a radius enclosing fixed overdensity (e.g. r_{200}).

Also, gas tracks dark matter within these halos. This behaviour would generate clusters with well-defined X-ray scaling relations.
For fixed redshift:

$\langle\rho\rangle \propto M/R^3$ is same for all systems

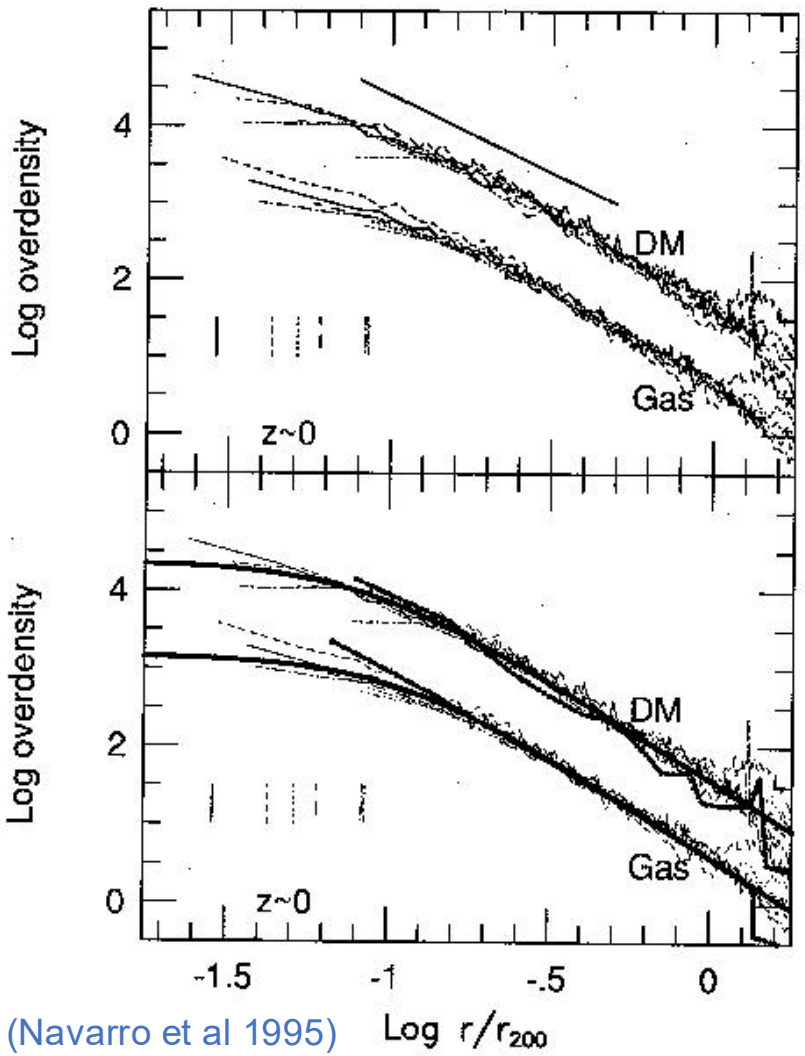
$T \propto M/R \propto R^2 \propto M^{2/3}$ from V.T.

$\therefore\ r_{200} \propto T^{1/2}$

$L_X \propto \rho^2 \cdot V \cdot \Lambda(T) \propto \rho^2 \cdot T^{3/2} \cdot \Lambda(T)$

where $\Lambda(T) \propto T^{1/2}$ for bremss,

$\rightarrow \mathbf{L_X \propto T^2}$

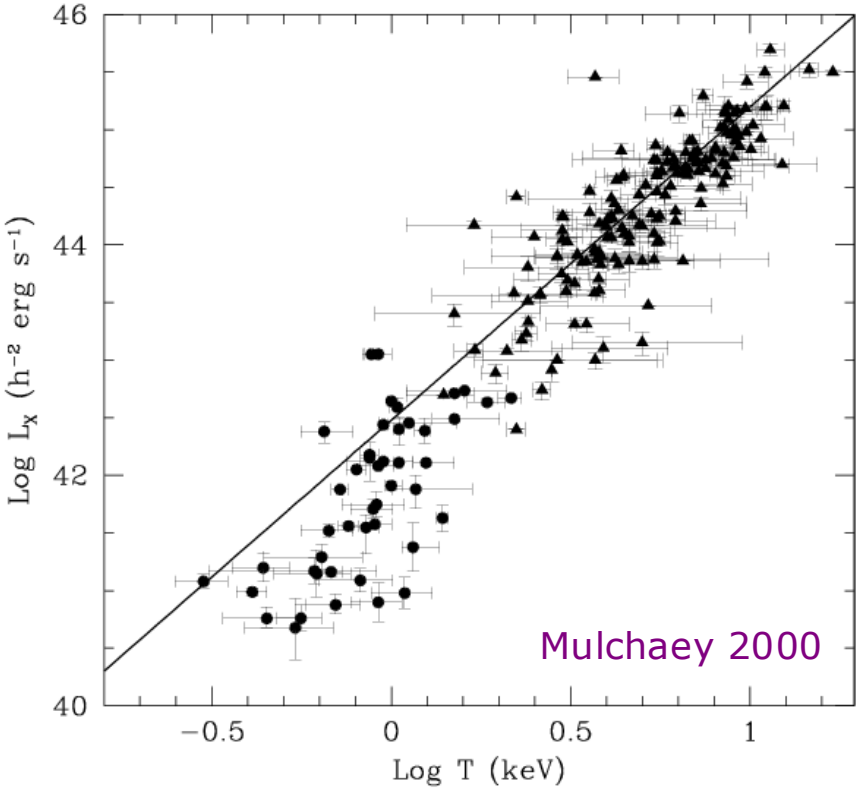


(Navarro et al 1995)

X-ray scaling properties

It has been clear for many years that the cluster L-T relation does not follow the $L \propto T^2$ slope expected for self-similar systems.

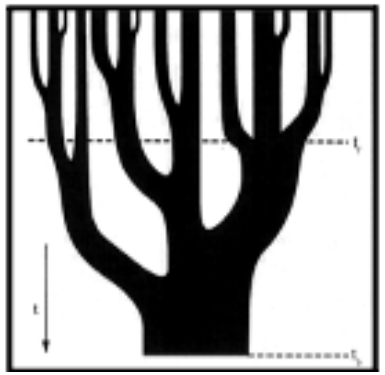
In practice, $L \propto T^{\sim 3}$ for clusters, with possible further steepening to $L \propto T^4$ in the group regime.



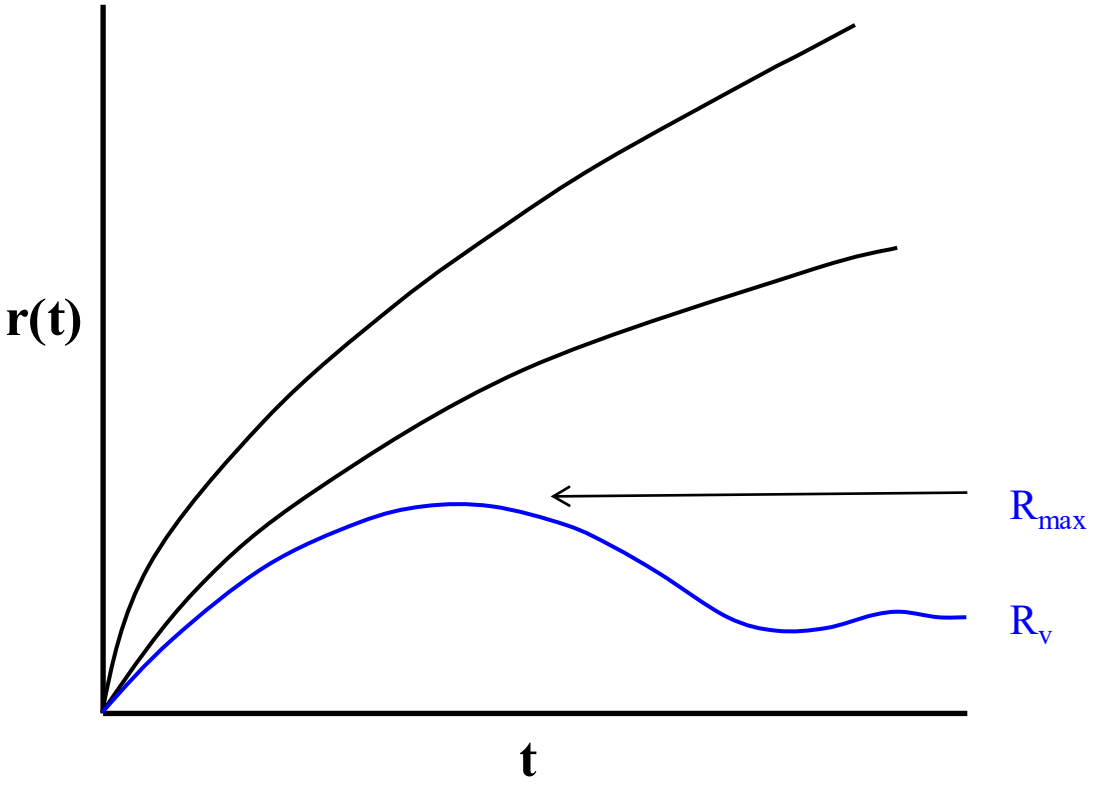
Clusters and Cosmology

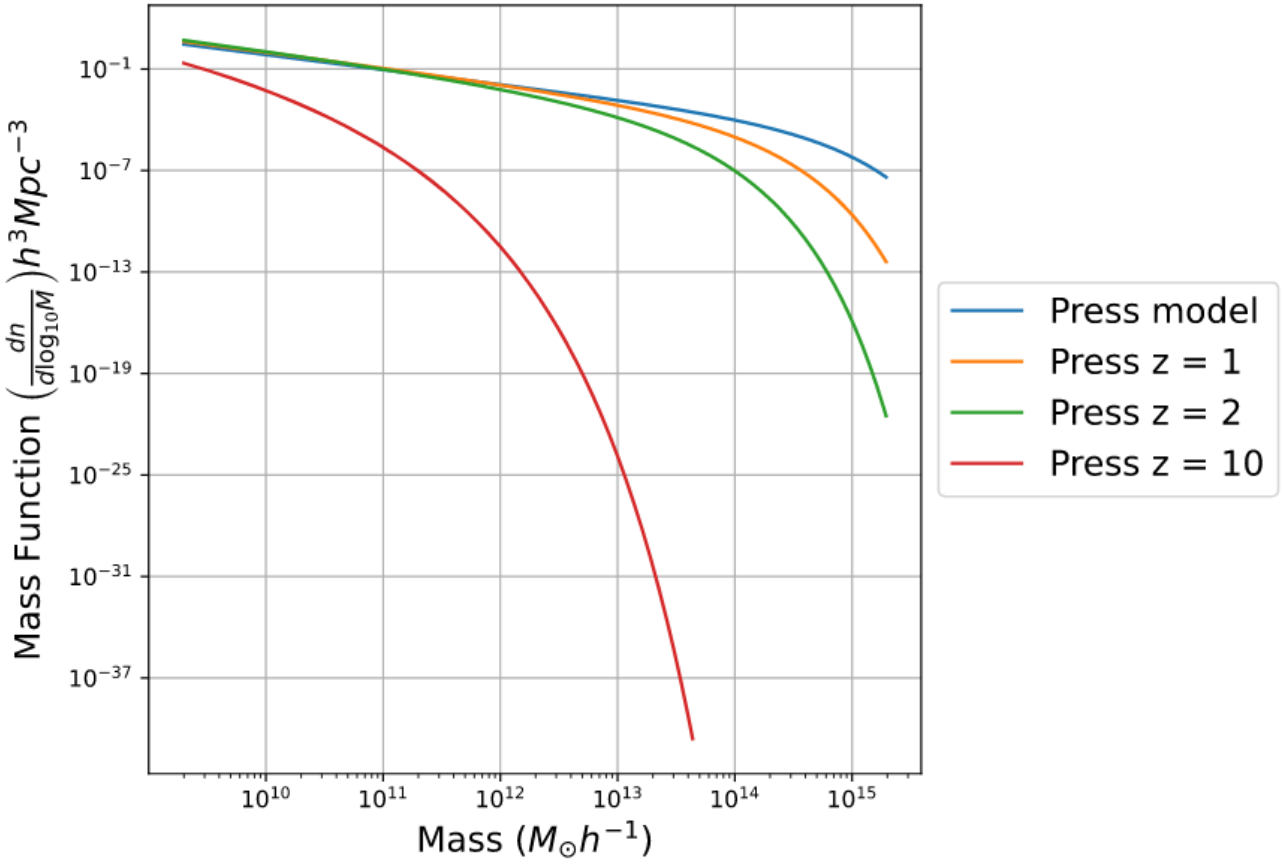
The halo mass function

We would like to know the numbers of halos of different masses which are collapsing out of the Hubble expansion at any given time.



Evolution of radii of concentric shells near a spherical overdensity.





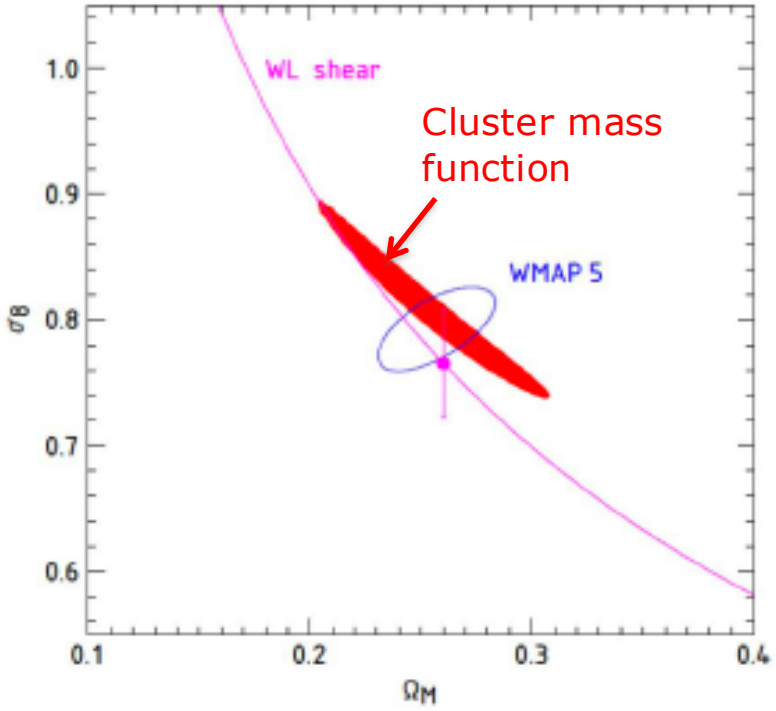
Cosmology with clusters

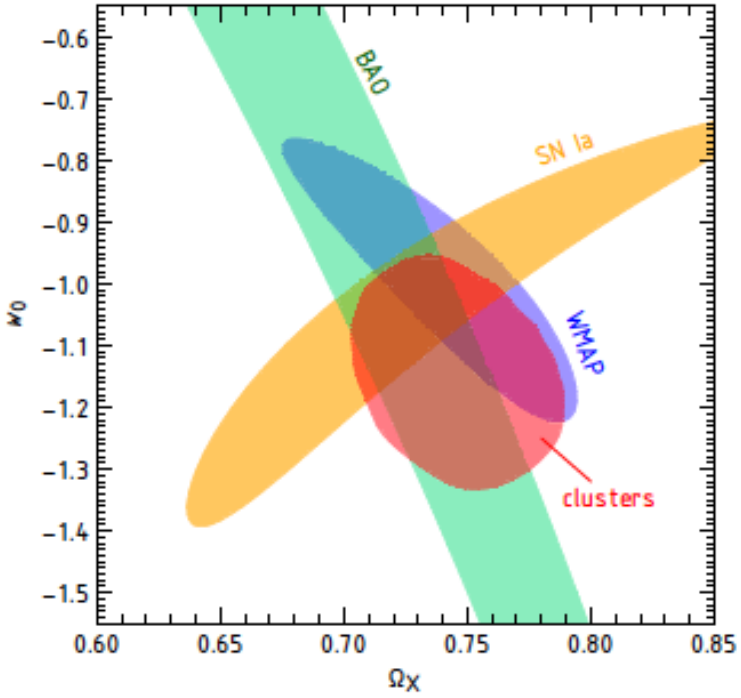
Observations of clusters can also be used to constrain cosmological models in two main ways:

a)The space density (mass function) of clusters and its evolution

b)The apparent gas fraction in rich clusters – assuming it is really constant

c)SZ measurement of – for instance – Hubble parameter





Vikhlinin et al 2009

Gas fraction cosmology

$$f_b = \frac{M_b}{M_{tot}} = \frac{M_{gas} + M_{stars}}{M_{tot}}$$

- ⋆ Measure M_{stars} from optical and M_{gas} from X-ray
- ⋆ Dominated by M_{gas} – measured value depends on d

Consider spherical region of cluster, angular radius θ

- ⋆ Physical (proper) radius of region is

$$R = \theta d_A$$

where d_A is angular diameter distance.

- ⋆ The proper volume of the region is just

$$V = \frac{4}{3}\pi(\theta d_A)^3$$

Gas fraction cosmology

Now M_{gas} is given by

$$M_{gas} = \rho_{gas} V$$

so

$$M_{gas} \propto \rho_{gas} (\theta d_A)^3$$

and recall that gas density is related to the X-ray luminosity

$$L \propto \int n_p n_e dV \propto \rho_{gas}^2 V$$

and we can relate L to the observed flux F:

$$L = F \times 4\pi d_L^2$$

where d_L is the luminosity distance to the cluster

Gas fraction cosmology

$$L \propto \int n_p n_e dV \propto \rho_{gas}^2 V \qquad\qquad L = F \times 4\pi d_L^2$$

Rearrange for ρ in terms of observables F and θ

$$\rho_{gas} \propto (L/V)^{1/2}$$

$$\rho_{gas} \propto \left(\frac{F d_L^2}{\theta^3 d_A^3}\right)^{1/2}$$

and finally

$$M_{gas} \propto \rho_{gas}(\theta d_A)^3 \propto (F\theta^3)^{1/2} d_L d_A^{3/2}$$

Gas fraction cosmology

$$M_{gas} \propto (F\theta^3)^{1/2} d_L d_A^{3/2}$$

For a cluster with observed F within angular radius θ

$$M_{gas} \propto d_L d_A^{3/2}$$

Can also show that if total mass is determined from X-ray hydrostatic analysis

$$M_{tot} \propto d_A$$

so the **estimated** value of the baryon fraction depends on distance as

$$f_{gas} = \frac{M_{gas}}{M_{tot}} \propto d_L d_A^{1/2}$$

Gas fraction cosmology

$$f_{gas} = \frac{M_{gas}}{M_{tot}} \propto d_L d_A^{1/2}$$

So to determine f_{gas} (or f_b) from X-ray observation of cluster at redshift z

⋆ need to know d_A, d_L to z

Instead, if we know f_b from theory, can determine d_A, d_L to z

⋆ constrain E(z)

Backup Slides

Gas fraction cosmology

Assuming that clusters large enough to be representative, expect f_b to be same at all z

- ⋆ at least until reach z clusters forming
- ⋆ "standard bucket"
- ⋆ observational determination of f_b depends on distance to cluster
  - sensitive to E(z)

$$\left(\frac{H(z)}{H_0}\right)^2 = \Omega_{\mathrm{m},0} \cdot (1+z)^3 + \Omega_{\mathrm{k},0} \cdot (1+z)^2 + \Omega_{\Lambda,0}$$

$$H(z) = H_0 \cdot E(z)^{1/2}$$

If we know f_b, and measure z, then infer distance to cluster

- ⋆ constrain E(z), hence $\Omega_M, \Omega_\Lambda, w$

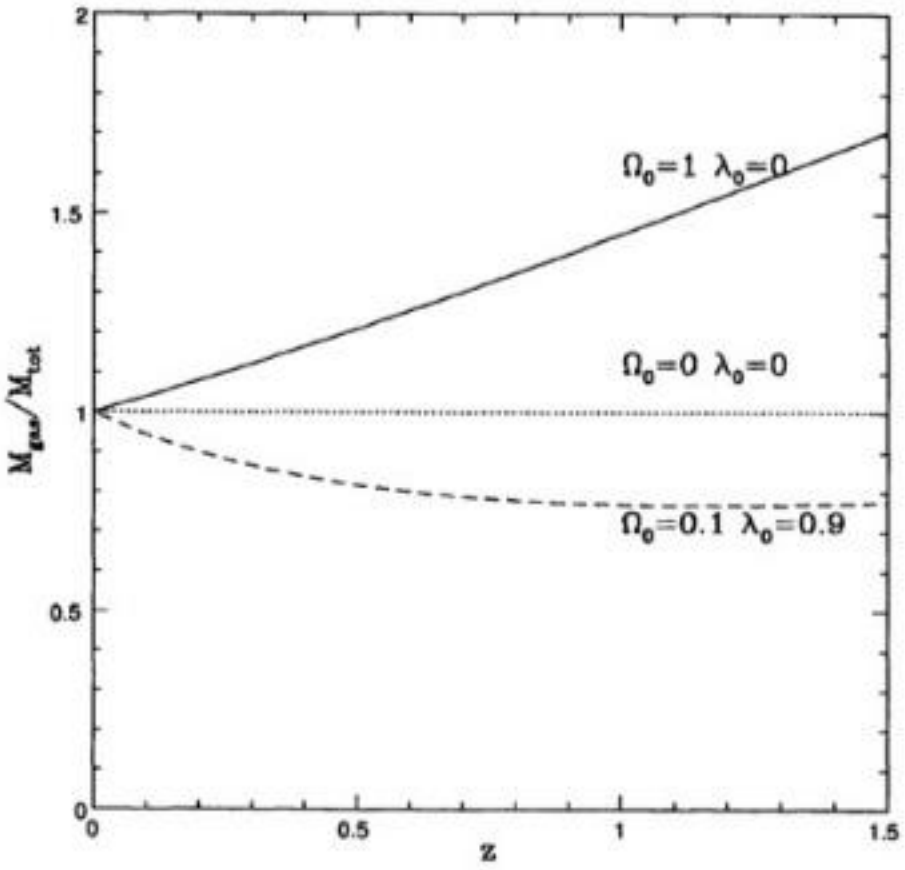
For strongest constraints, measure f_b for set of clusters at different z

Gas fraction cosmology

Method first proposed by Sasaki (1996, PASJ, 48)

Artificial evolution in f_{gas} for different cosmologies

Only get constant f_{gas} with z if using correct cosmology



Gas fraction cosmology

$$f_b = \frac{M_b}{M_{tot}} = \frac{M_{gas} + M_{stars}}{M_{tot}}$$

- ⋆ Measure M_{stars} from optical and M_{gas} from X-ray
- ⋆ Dominated by M_{gas} – measured value depends on d

Consider spherical region of cluster, angular radius θ

- ⋆ Physical (proper) radius of region is

$$R = \theta d_A$$

where d_A is angular diameter distance.

- ⋆ The proper volume of the region is just

$$V = \frac{4}{3}\pi(\theta d_A)^3$$

Gas fraction cosmology

Now M_{gas} is given by

$$M_{gas} = \rho_{gas} V$$

so

$$M_{gas} \propto \rho_{gas} (\theta d_A)^3$$

and recall that gas density is related to the X-ray luminosity

$$L \propto \int n_p n_e dV \propto \rho_{gas}^2 V$$

and we can relate L to the observed flux F:

$$L = F \times 4\pi d_L^2$$

where d_L is the luminosity distance to the cluster

Gas fraction cosmology

$$L \propto \int n_p n_e dV \propto \rho_{gas}^2 V \qquad\qquad L = F \times 4\pi d_L^2$$

Rearrange for ρ in terms of observables F and θ

$$\rho_{gas} \propto (L/V)^{1/2}$$

$$\rho_{gas} \propto \left(\frac{F d_L^2}{\theta^3 d_A^3}\right)^{1/2}$$

and finally

$$M_{gas} \propto \rho_{gas}(\theta d_A)^3 \propto (F\theta^3)^{1/2} d_L d_A^{3/2}$$

Gas fraction cosmology

$$M_{gas} \propto (F\theta^3)^{1/2} d_L d_A^{3/2}$$

For a cluster with observed F within angular radius θ

$$M_{gas} \propto d_L d_A^{3/2}$$

Can also show that if total mass is determined from X-ray hydrostatic analysis

$$M_{tot} \propto d_A$$

so the **estimated** value of the baryon fraction depends on distance as

$$f_{gas} = \frac{M_{gas}}{M_{tot}} \propto d_L d_A^{1/2}$$

Gas fraction cosmology

$$f_{gas} = \frac{M_{gas}}{M_{tot}} \propto d_L d_A^{1/2}$$

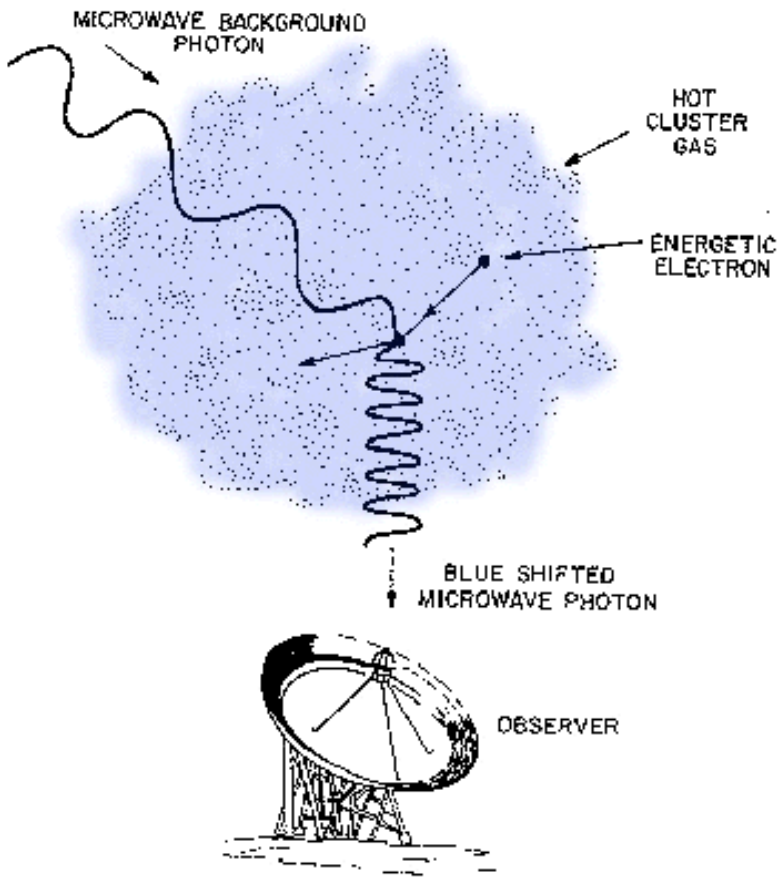
So to determine f_{gas} (or f_b) from X-ray observation of cluster at redshift z

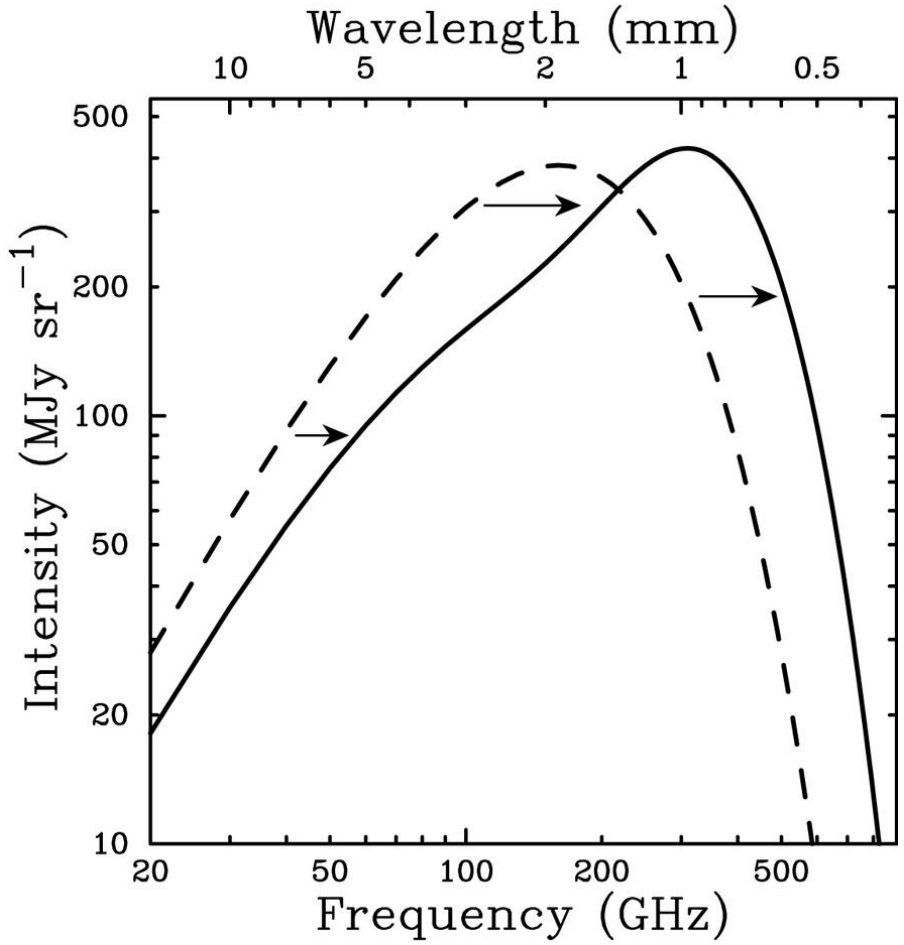
⋆ need to know d_A, d_L to z

Instead, if we know f_b from theory, can determine d_A, d_L to z

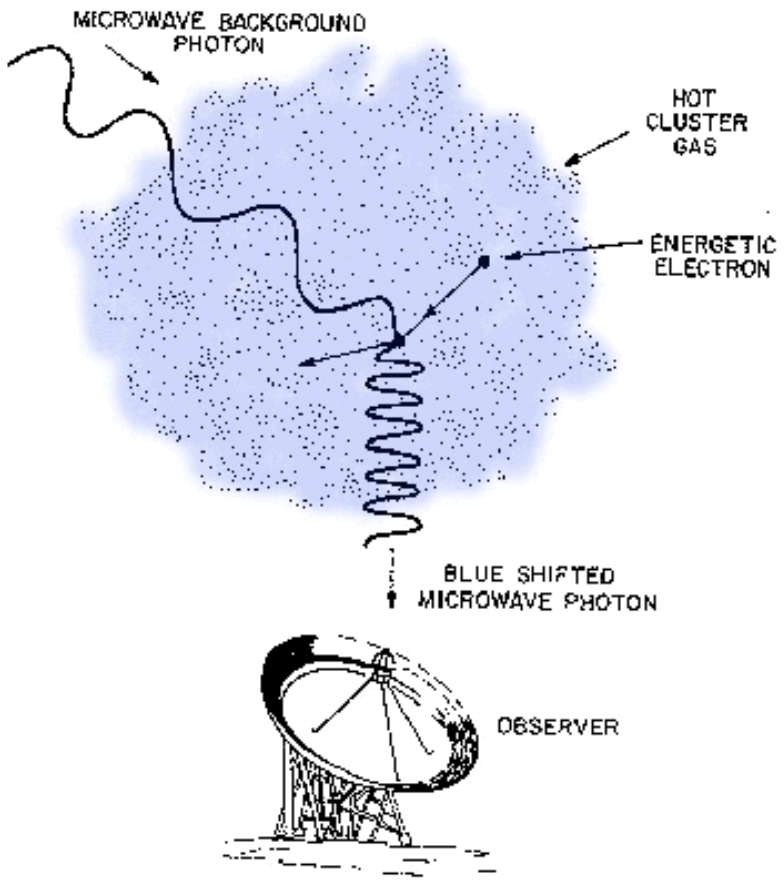
⋆ constrain E(z)

SZ measurement of H0





SZ measurement of H0



The y-parameter measures the energy transfer from hot gas in galaxy cluster to CMB photons from inverse Compton scattering. This is given by:

$$y = \int \frac{k_B T_e}{m_e c^2} \, n_e \sigma_T dl$$

Let's say the cluster is an isothermal sphere with a density profile of

$$n_e = n_{e,0} \left(1 + \frac{r^2}{r_c^2}\right)^{-3/2}$$

Then integrating over the line of sight of a given impact parameter b is

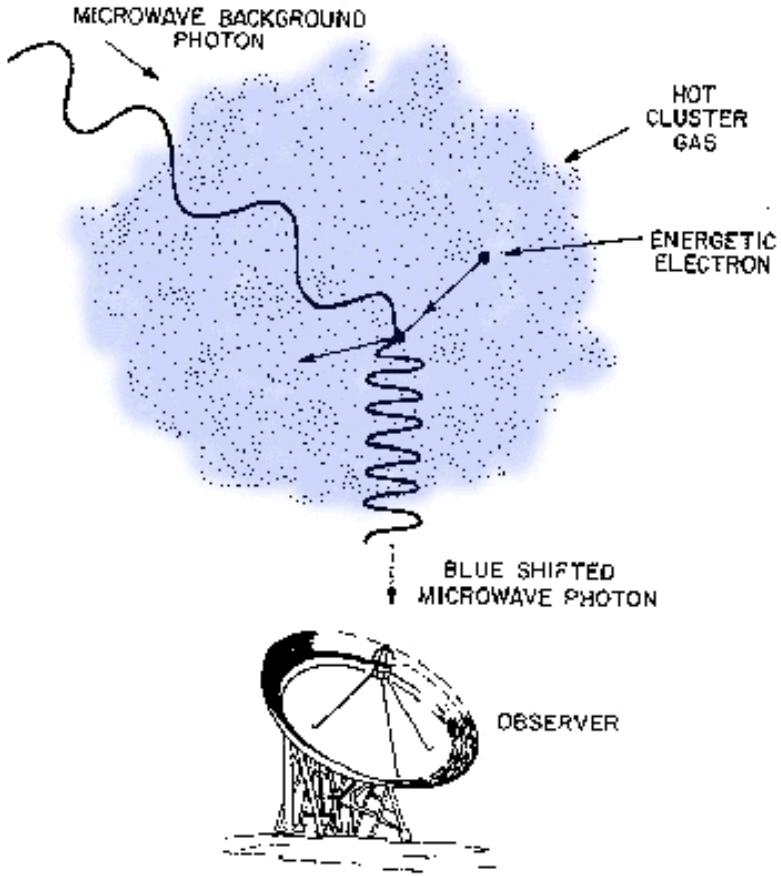
$$y = \frac{k_B T_e}{m_e c^2} \, \sigma_T \int_{-\infty}^{\infty} n_{e,0} \left(1 + \frac{b^2 + l^2}{r_c^2}\right)^{-3/2} dl$$

Where $r^2 = b^2 + l^2$ so, at the cluster center:

$$y_0 = \frac{k_B T_e}{m_e c^2} \, \sigma_T \, n_{e,0} \, r_c$$

The y is proportional directly to the thermal energy of the ICM.

SZ measurement of H0



This y parameter can be integrated over the whole cluster to give Y.

$$Y = \int y \, d\Omega \propto \int n_e T_e dV$$

The electron number density is

$$N_e = \frac{M \, f_b}{\mu m_p}$$

Since the temperature follows from the virial equilibrium:

$$T_e \propto \frac{GM}{R} \propto M^{\frac{2}{3}} E(z)^{\frac{2}{3}}$$

So, in total, this becomes

$$Y \propto M^{\frac{5}{3}} E(z)^{-2/3}$$

The integrated flux density of the cluster depends only on the intrinsic properties of the cluster and the angular diameter distance. Thus, the Hubble parameter can be directly measured.

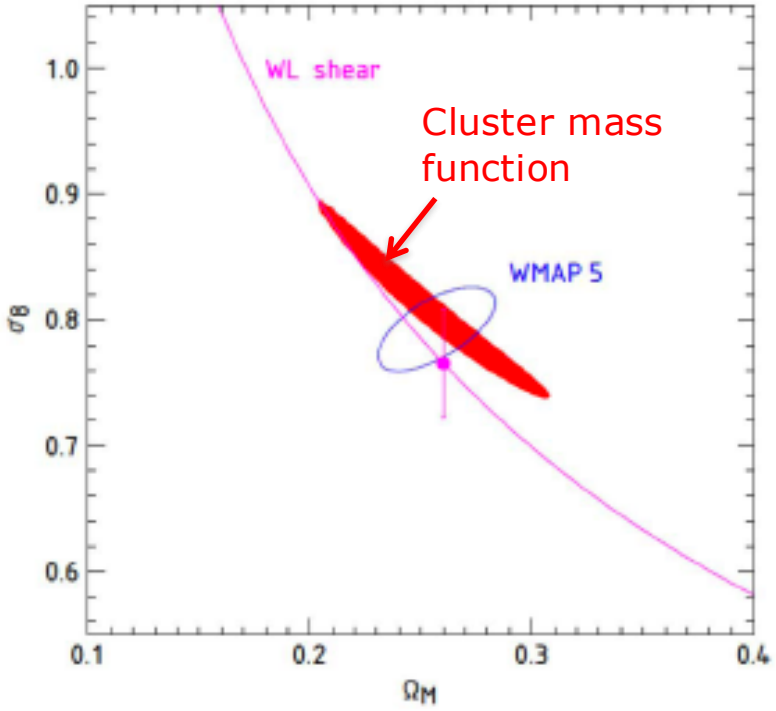
Cosmology with clusters

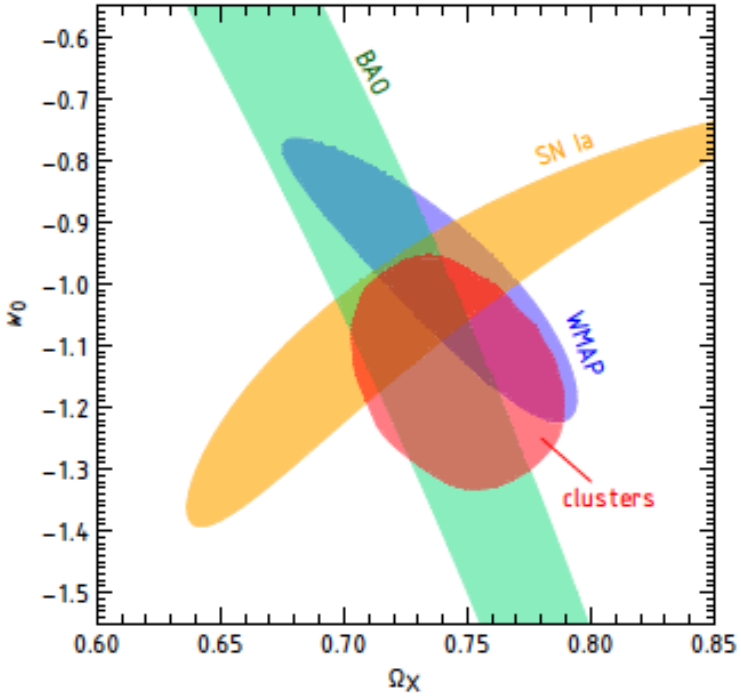
Observations of clusters can also be used to constrain cosmological models in two main ways:

a)The space density (mass function) of clusters and its evolution

b)The apparent gas fraction in rich clusters – assuming it is really constant

c)SZ measurement of – for instance – Hubble parameter





Vikhlinin et al 2009

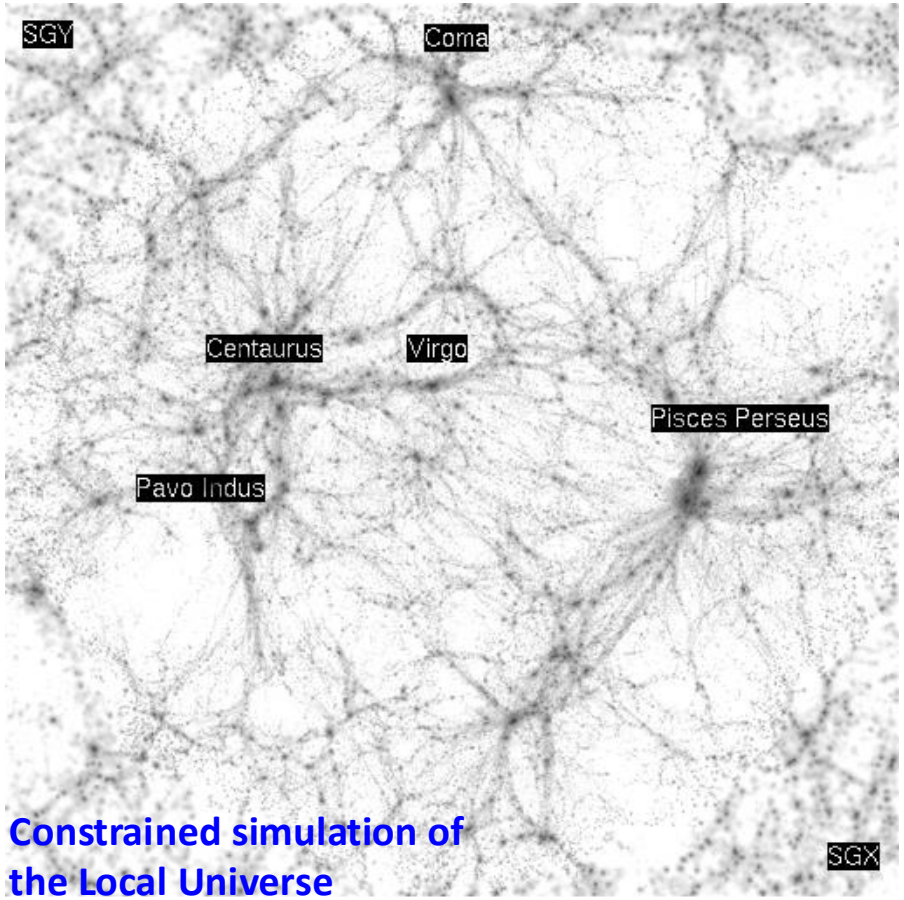
§ 9 – The cosmic web

How are galaxies distributed in the Universe, and how can this distribution be characterised statistically? What is the state of the intergalactic gas which contains the majority of the baryons and how has this evolved since reionization?

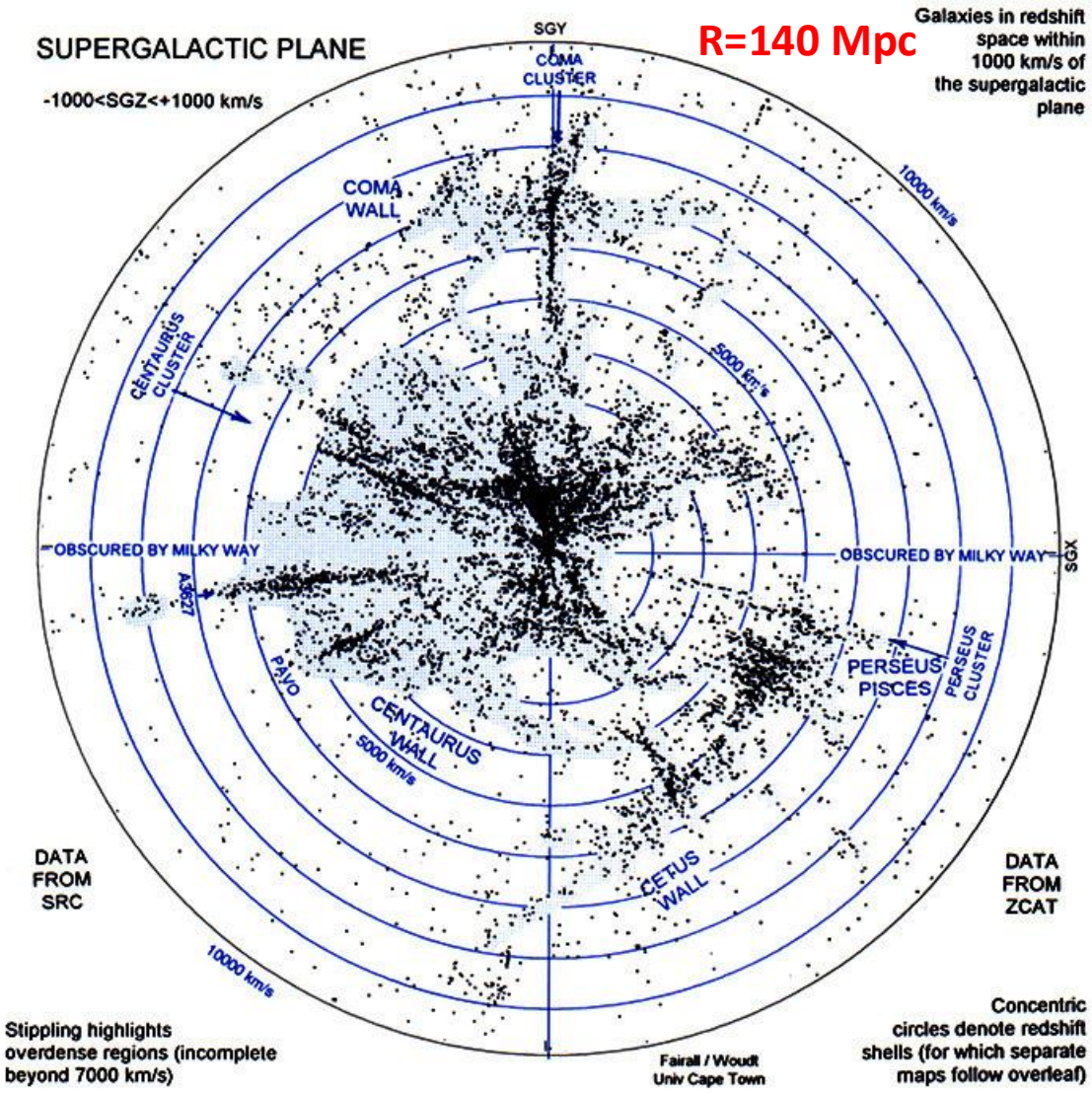
- The structure of the local Universe
- Quantifying clustering

The local Universe

The supergalactic plane extends further, to a distance >100 Mpc, and contains a number of large galaxy clusters. Note the "finger of God" effect, which arises due to…



Constrained simulation of the Local Universe



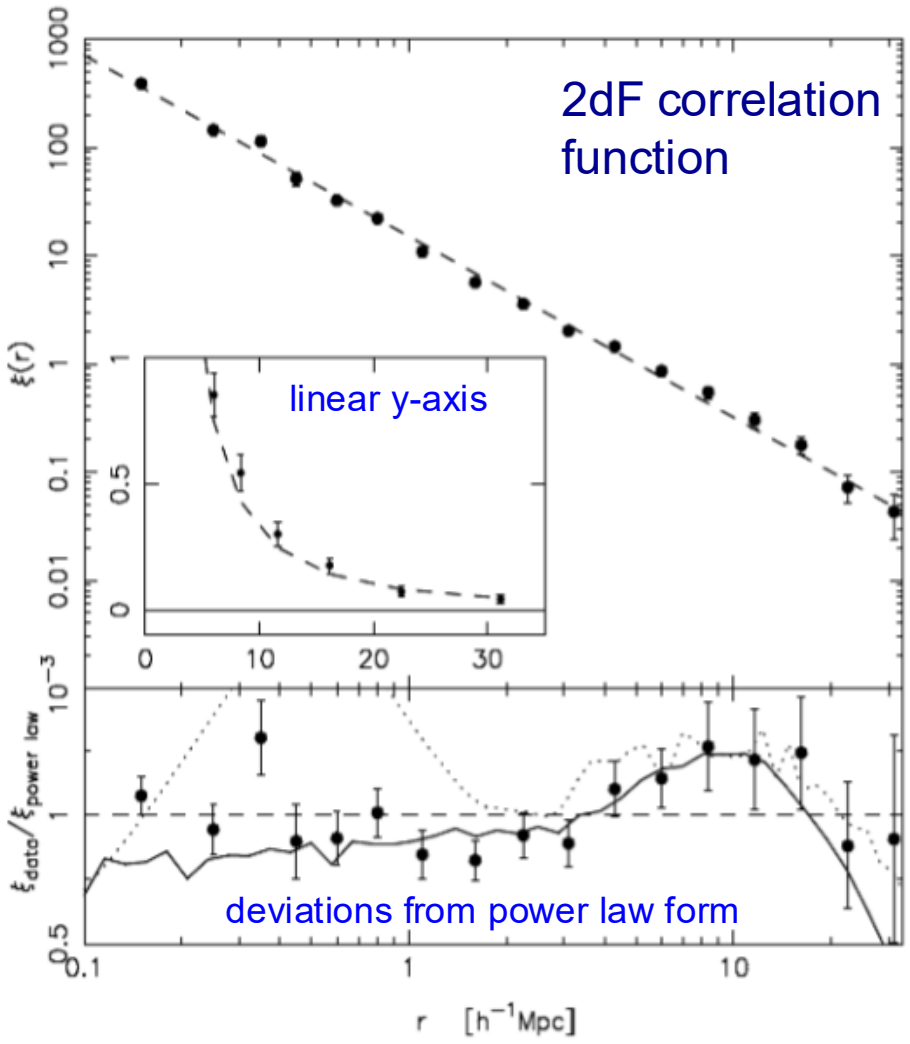
Quantifying clustering

We know that galaxies are not randomly scattered. The degree of clustering can be quantified using the two-point correlation function $\xi(r)$. If the mean galaxy density is n, then the joint probability of finding a galaxy in each of two small volumes V_1 and V_2 separated by distance r_{12} is

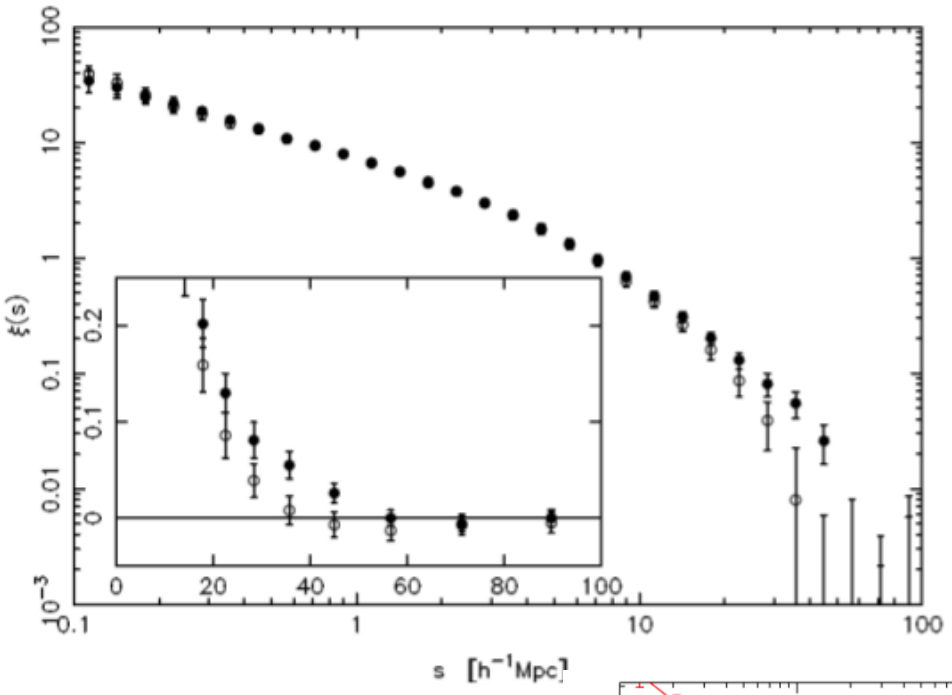
$$P_{12} = n^2 [1+\xi(r_{12})] V_1 V_2 .$$

This would simply reduce to $n^2 V_1 V_2$ in the absence of any clustering, so $\xi(r_{12})$ represents the excess probability arising from the clumping of galaxies. Clearly it is dimensionless. If $\xi(r_{12})>0$ at small r_{12} then galaxies are clumped. If $\xi(r_{12})<0$ then they avoid one another.

Typically, it is found to be approximately power law at small separations $\xi(r)=(r/r_0)^{-\gamma}$, where typically $r_0 \approx 7$ Mpc and $\gamma \approx 1.7$

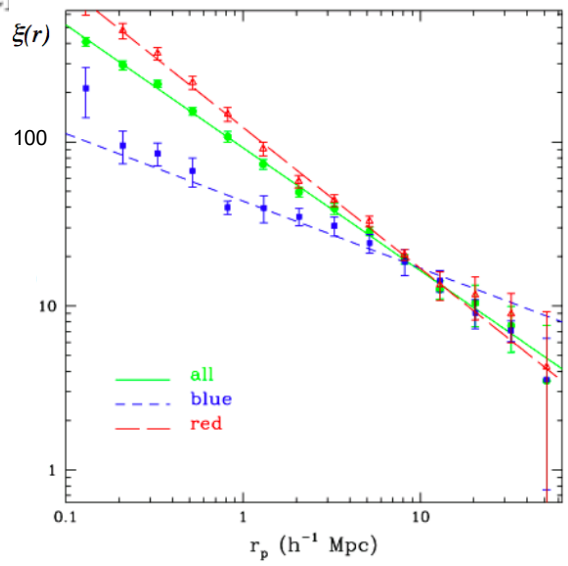


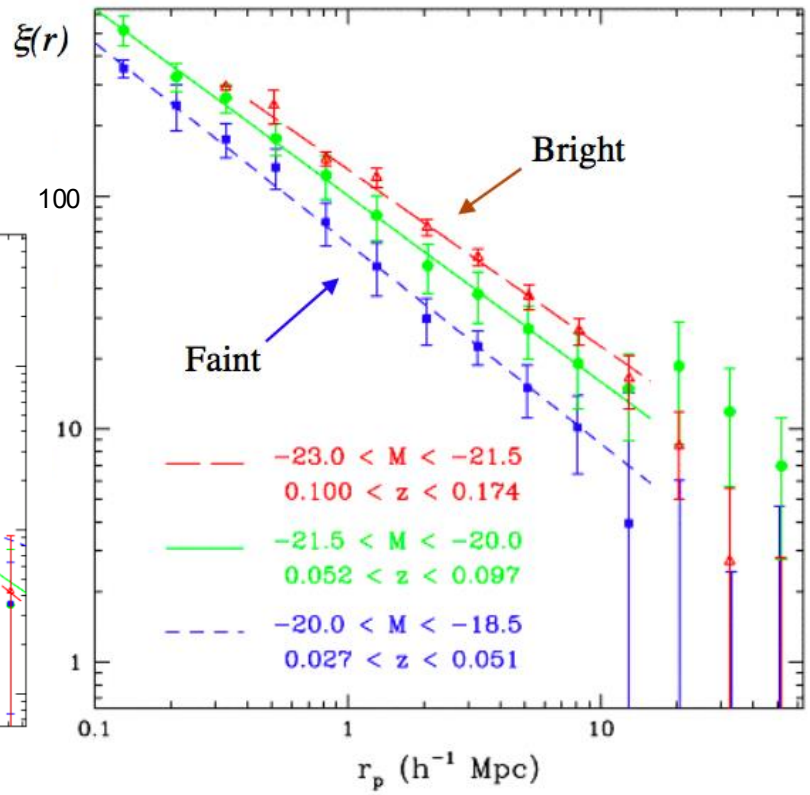
Quantifying clustering



On scales comparable with the sizes of voids, sheets and filaments ξ(r) drops to zero, indicating that on larger scales the Universe is essentially homogeneous.

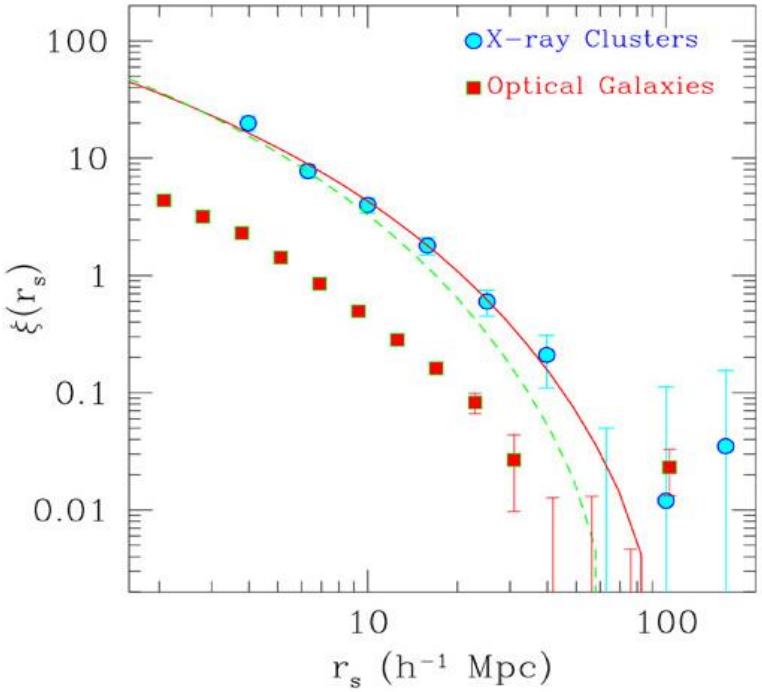
Bright galaxies are found to be more strongly clustered than dim ones, and early-types more than late-types.

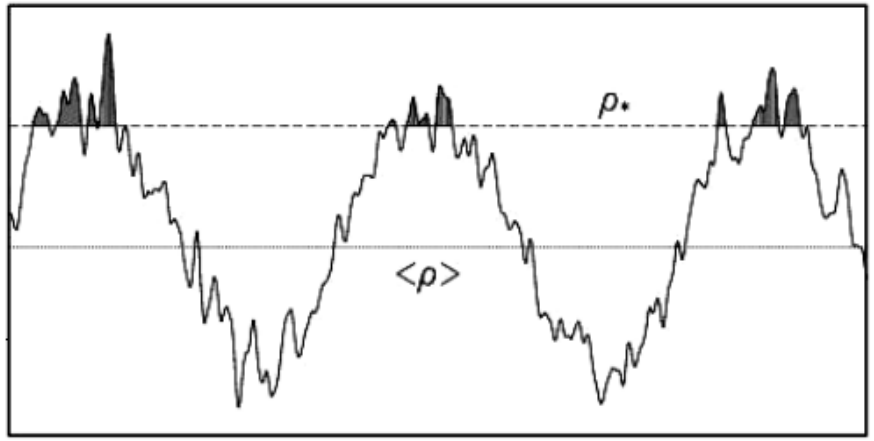




Quantifying clustering

Even larger structures, such as groups and clusters are even more strongly clustered, with larger values of the correlation length r_0 , which may be as high as 40 Mpc for rich clusters.



Since different systems are clustered to different degrees, it is clear that they cannot all be following the dark matter distribution – i.e. they are biased tracers of the mass distribution.

One way this can be understood is in terms of rare peaks. Since perturbations turn around and collapse when they reach a critical density, it follows that the first objects to collapse are likely to be those sitting on top of larger scale perturbations. This will cause them to cluster strongly.